Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) First Year, First Semester Final Examination-2022

Course Code: EEE 0713 1121 **Session:** 2020-2021

Course Title: Course Title For This Course Code

Total Marks: 60 **Credits:** 3.0 **Total Time:** 3 Hours

[The figures in the right margin indicate full marks]

Part-A

*[Answer **all** the questions]*

1. (a) What are luminescent and photo-luminescent processes? 2

   (b) How is the concept of quasi-Fermi level useful in semiconductors? 2

   (c) If you want the marks to be displayed as $2 + 2$, use the command `\bdpart[2+2]` instead of the regular `\part[]` command. As for example: What is molecular orbital? Explain bonding and anti-bonding in molecular orbital. 2+2

   (d) If you want the subparts of the question to be appeared inline the use the command `\begin{inlinesubparts} \item, \item, etc.---\end{inlinesubparts}`. As for example: Draw schematic diagrams to show splitting of two molecular orbitals due to resonance interactions with the orbitals: (i) having same energy, and (ii) having different energies. 2+2

2. (a) For units, you can use SI units; for example, 2 µm can be achieved by using the command `\qty{2}{\micro\meter}` or `\SI{2}{\micro\meter}`. This type `\qty{2}{\micro\meter}` is the modern standard in `siunitx` v3, while `\SI{2}{\micro\meter}` is the legacy syntax from older versions. 2

   (b) In case you need to use the unit for an expression as for example $20sin(\omega t)$ mA, right after the expression, you can use the command `\si{\milli\ampere}`. 1

   (c) Two resistors of values 1 kΩ and 4 kΩ are connected in series across a constant voltage supply of 100 V. A voltmeter having an internal resistance of 12 kΩ is connected across the 4 kΩ resistor. Draw the circuit and calculate: 4

   (i) True voltage across 4 kΩ resistor before the voltmeter was connected.

   (ii) Actual voltage across 4 kΩ resistor after the voltmeter is connected and voltage recorded by the voltmeter.

   (d) Figure(pdf, png, etc) can be inserted using the `\includegraphics{nameOfFigure}` command as shown below. : Find the rms value of the current waveform of Fig.1. If the current flows through a 9 Ω resistor, calculate the average power absorbed by the resistor. 5

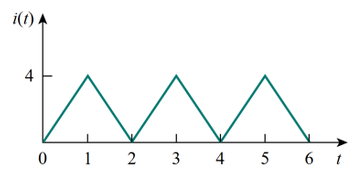


Fig. 1

3. (a) If you want to place two figures side by side then use minipage environment. As for example: Using nodal analysis, determine the potential across the 4 Ω resistor in Fig. 2 4

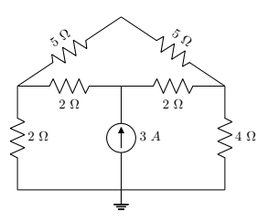


Fig. 2

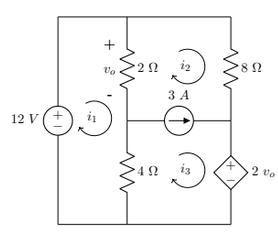


Fig. 3

   (b) Use mess analysis to find currents and voltage v_o in the circuit of Fig. 3. 4

(c) If a table is needed to insert, then you can use either tabular or tabulrx. In the table as shown in Table 1, the data are given for a general purpose Silicon diode, draw the I-V characteristic curve. 2

Table 1: Your Table Title Here

SL.no	Forward bias voltage (V)	Forward bias current(mA)
1	0	0
2	0.2	0.0
3	0.4	0.1
4	0.5	0.5
5	0.53	1.0
6	0.6	8.2
7	0.66	19.5
8	0.7	53.5
9	0.71	83.1
10	0.73	112.7

OR

(a) Find the Thévenin's theorem equivalent circuit of Fig. 4 to the left of the terminal. 4

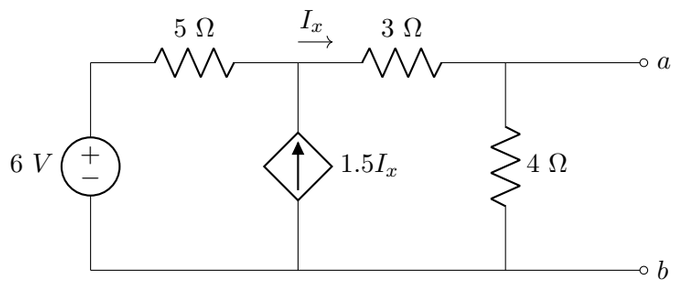


Fig. 4

(b) Find the magnitude R_L for the maximum power transfer in the circuit shown in Fig. 5. Also find out the maximum power. 4

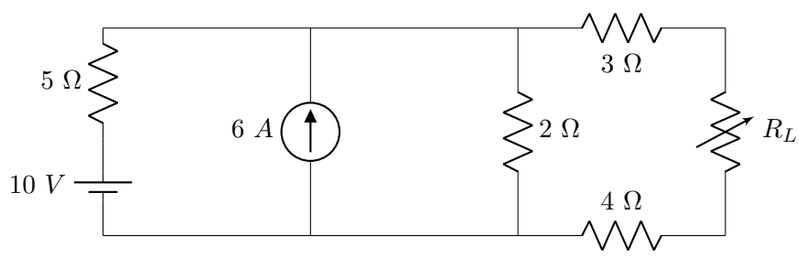


Fig. 5

(c) Write short notes on Real power and Reactive power. 2

Part-B

*[Answer **all** the questions]*

4. (a) An Inverse Definite Minimum Time (IDMT) overcurrent relay is connected to a 400/5 A Current Transformer (CT) with a Plug Setting (PS) of 125%. The characteristic curve for TMS = 1.0 is shown in Fig. 6. If a symmetrical fault current of 5000 A flows through the primary side, calculate: 6
   (i) The Plug Setting Multiplier (PSM) of the relay.
   (ii) The operating time of the relay for TMS = 0.4.
   (iii) The required TMS setting if the relay must operate in 1.8 seconds.

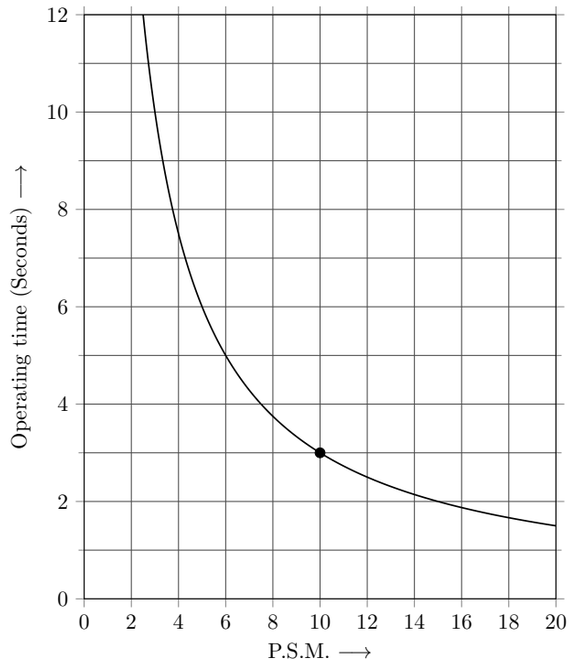


Fig. 6: IDMT Overcurrent Relay Characteristic Curve

(b) For the emitter bias network of Fig. 7 determine I_B, I_C, V_{CE}, V_C, V_E and V_B. 4

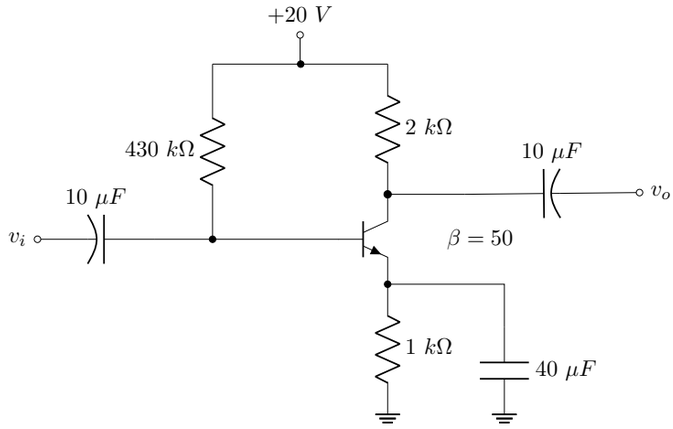


Fig. 7

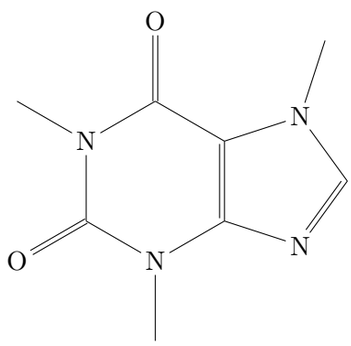
5. (a) Using the command `\ce{..}` you can typeset the chemical formula and equations. As for example: Write down whether the given chemical formulas are organic or inorganic. 2

- Water: `\ce{H2O}`, H_2O
- Benzene: `\ce{C6H6}`, C_6H_6
- Hydrogen peroxide: `\ce{H2O2}`, H_2O_2
- Acetic acid: `\ce{C2H4O2}`, $C_2H_4O_2$
- Glucose: `\ce{C6H12O6}`, $C_6H_{12}O_6$

(b) The chemical equation can also be written using the command `\ce{}`. As for example: Explain the following two chemical equations. 2+2

- `\ce{2H2 + O2 -> 2H2O}` typesets $2\,H_2 + O_2 \longrightarrow 2\,H_2O$
- `\ce{CO2 + C -> 2 CO}` typesets $CO_2 + C \longrightarrow 2\,CO$

(c) Drawing a molecule consists mainly of connecting groups of atoms with lines. Simple linear formulae can be easily drawn using the chemfig package and using the command `\chemfig{*6((=O)-N(-)-(*5(-N=-N(-)-))=-(=O)-N(-)-)}`, as shown in the following example: Identify the given chemical formula. 1+4



6. (a) Simplify the Boolean function $F(w, x, y, z) = \sum(1, 3, 7, 11, 15)$. Which has don't-care condition: $d(w, x, y, z) = \sum(0, 2, 5)$. 4

(b) Simplify $F(A, B, C, D) = \sum(0, 1, 2, 5, 8, 9, 10)$ in product of sums. 4

(c) Define Minterms and Maxterms and briefly explain De Morgan's law. 2

OR

(a) Simplify the Boolean function $F(w, x, y, z) = \sum(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$. 4

(b) A transmission line has the following per-unit length parameters: $L = 0.5\,\mu H\,m^{-1}$, $C = 200\,pF\,m^{-1}$, $R = 4\,\Omega\,m^{-1}$ and $G = 0.02\,S\,m^{-1}$. Calculate the propagation constant and characteristics impedance of this line. If the line is 30 cm long, what is the attenuation in dB. 4

(c) If you only want the unit as symbol mode like $R = 4\,\Omega/m$ instead of the conventional unit $R = 4\,\Omega\,m^{-1}$, wrap the unit in braces: as shown below. 2½

```
{
\sisetup{per-mode = symbol}
\si{\milli\volt\per\pico\second}
}
```

Table of Specification for the Summative Assessment (Semester Final Exam)

Department of Electrical and Electronic Engineering

B.Sc. (Engg.) First Year, First Semester Final Examination-2022

Course Code: EEE 0713 1121 **Session:** 2020-2021

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Q. No.	Marks	CO No.	Bloom's Levels of Cognition					
			Remember	Understand	Apply	Analyze	Evaluate	Create
1a	3	CO1	2	1				
1b	2	CO1	1	1				
1c	5	CO1	3	2				
2a	3	CO1	1	1	1			
2b	3	CO1	2	1				
2c	4	CO1	2	1	1			
3a	4	CO2		2	2			
3b	6	CO2		2	2	1	1	
4a	4	CO3	2	2				
4b	4	CO3		2	1	1		
4c	2	CO3				1	1	
5a	4	CO4		1	1	1		1
5b	6	CO4			2	2	2	
6a	2	CO5			1	1		
6b	5	CO5				3	1	1
6c	3	CO5	1	1	1			
Total	**60**		**14**	**17**	**12**	**10**	**5**	**2**
Total % in each level			**23.33**	**28.33**	**20.00**	**16.67**	**8.33**	**3.33**
% Marks in Major levels			**71.67**			**28.33**		

SUST, Sylhet, Bangladesh
October 21, 2026

Signature

Name of question setter